simple but highly scalable—throughput and storage—message queues that guarantee the delivery of messages at least once with the following characteristics:

- Under rare circumstances, a single message will be available for consumption twice. This may sound strange if you compare it to other message queues, but you'll see how to deal with this problem later in the chapter.
- SQS doesn't guarantee the order of messages, so you may read messages in a different order than they were produced. Learn more about the message order at the end of this section.

This limitation of SQS is also beneficial for the following reasons:

- You can put as many messages into SQS as you like.
- The message queue scales with the number of messages you produce and consume.
- SQS is highly available by default.
- You pay per message.

The pricing model is simple: $0.24 to $0.40 USD per million requests. Also, the first million requests per month are free. It is important to know that producing a message counts as a request, and consuming is another request. If your payload is larger than 64 KB, every 64 KB chunk counts as one request.

We have observed that many applications default to a synchronous process. That's probably because we are used to the request-response model and sometimes forget to think outside the box. However, replacing a synchronous with an asynchronous process enables many advantages in the cloud. Most importantly, scaling